

Topographical Orthogonal Generative Theory (TOGT): A Computable Higher-Order Logic and Category-Theoretic Framework

Pablo Grossi

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Abstract

We present TOGT, a computable higher-order logic and category-theoretic framework unifying generative processes across domains. The core is a four-operator algebra (Compression, Constraint, Folding, Unfolding) organized in an algorithmically closed pipeline $S(L) \rightarrow \Phi \rightarrow \mathbf{g} \rightarrow \mathbf{K}(L) \rightarrow \text{Space}$. We formalize $\mathbf{g} = \nabla\Phi$, its sixfold composition $\mathbf{g}^6$ with a testable invariant (33 under constraints), and outline $\mathbf{g}^{64}$ (Kether Orthogon) as a 4-functor yielding a 4D crystalline spine.

1 Background

Complex systems exhibit directional, constrained, recursive generation. TOGT supplies a computable logic and category semantics to map patterns across scales.

2 Framework

Sorts: Top, Stat, Dyn; potentials Φ , flows $\mathbf{g} = \nabla\Phi$, curvature $\mathbf{K}$; operators $\{\textit{Comp}, \textit{Konst}, \textit{Fold}, \textit{Unfold}\}$. Aims include formal proofs, software, and validations across domains.

3 Minimal Model

$$\Phi(x; \lambda) = \frac{1}{4}x^4 - \frac{1}{2}\lambda x^2, \dot{x} = \lambda x - x^3.$$

4 Notes

See [1, 2] for context; toolchains include TTK [3].

References

- [1] D. Castelvechi. Microsoft claims quantum-computing breakthrough—but some physicists are sceptical. *Nature*, 2025.
- [2] M. M. S. Ismail et al. Early Warning Signals of Financial Crises Using Persistent Homology. *Frontiers in Applied Mathematics and Statistics*, 2022.
- [3] Topology ToolKit (TTK). Project site and repository, 2019–2026.